

Chanukya Chilamkuri

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PROFESSIONAL SUMMARY

Forward Deployed Engineer with 3+ years of production software engineering experience designing, building, and shipping agentic AI and full-stack systems directly for end customers. Proven track record owning technical components end-to-end, from architecture and prompt design through deployment, validation, and production observability, including LLM-powered RAG agents, autonomous tool-use systems, and cross-platform mobile architectures. Experienced translating ambiguous business requirements gathered directly from non-technical stakeholders into deployed, customer-facing solutions. Available for immediate, full-time roles (completing MS Computer Science (remote), graduating Dec 2026).

TECHNICAL SKILLS

- **AI/ LLM:** RAG Pipelines, Prompt Engineering, Multi-step Planning, Tool Use, Memory Systems, Agent Evaluation, LangChain, LlamaIndex, OpenAI API, Anthropic API, pgvector, Vector Databases
- **Languages & Backend:** Python, TypeScript, JavaScript, Java, C++, FastAPI, Flask, Node.js, Express, REST API Design, WebSockets, RBAC, SQL
- **Data & Integration:** Large-Scale Data Preprocessing, Data Modelling & Governance, PostgreSQL, MongoDB, Firebase Firestore, MySQL
- **Cloud & DevOps:** AWS (EC2, S3, Lambda), Docker, CI/CD Pipelines (GitHub Actions, Vercel, Railway), Git/GitHub, Azure, Production Deployments & Observability
- **Frontend & Mobile:** React, React Native, Expo, Tailwind CSS, Redux Toolkit

PROFESSIONAL EXPERIENCE

Next Play

Apr 2026 - Jul 2026

Forward Deployed Engineer Intern | Agentic Mobile Systems

- Shipped 0-1 production-ready architectures by gathering ambiguous requirements directly from non-engineering stakeholders, rapid-prototyping features, and driving iterations directly based on customer feedback.
- Accelerated real-time delivery of context-aware alerts across iOS and Android by architecting and deploying an end-to-end cross-platform push notification agent utilizing APNs and FCM routing.
- Reduced notification latency and improved user context retention by implementing multi-step orchestration with quiet-hours enforcement and LLM-style memory systems to maintain session states.
- Streamlined feature-to-production velocity ("spec to App Store") by delivering the entire Coach IQ module (XP tracking, marketplace schema, video fallback, mobile UI) using React Native, TypeScript, and Node.js.
- Ensured 99.9% deployment reliability by building and maintaining automated CI/CD pipelines via GitHub Actions for mobile builds and backend services.

Morningstar

Oct 2022 - May 2025

Full-Stack Software Engineer

- Cut average lookup times by 40% for internal stakeholders by designing and deploying a production LLM-powered RAG agent over 500+ internal documents, using Lakehouse architecture principles to clean, unify, and ingest unstructured data streams into pgvector via FastAPI.
- Boosted RAG output quality and reduced hallucination rates by engineering a custom evaluation framework to benchmark prompt instructions, model reasoning, and factual grounding.
- Enabled multi-step autonomous actions with contextual memory by integrating tool-use workflows connecting core models to internal APIs, knowledge bases, and databases using LangChain and LlamaIndex.
- Streamlined internal ML model deployments by designing distributed system architectures on AWS (EC2, S3, Lambda) with robust automated CI/CD pipelines.
- Drove product adoption for thousands of active users by shipping high-performance full-stack features using React, TypeScript, and Python with secure, role-based access control (RBAC).
- Accelerated team delivery speed by mentoring junior engineers, running technical design reviews, and authoring comprehensive system architecture documentation.

PROJECT EXPERIENCE

Ask My Resume: Agentic RAG System

Mar 2026 - Apr 2026

Personal Project | [Link](#)

- Built a production-grade 0-1 agentic RAG system featuring semantic chunking, OpenAI embeddings, PostgreSQL + pgvector ANN search, top-k retrieval, and prompt-engineered context injection using Claude and GPT-4o.
- Guaranteed answer accuracy and factual grounding across 50+ complex queries by designing production prompts with tight guardrails and building automated evaluation tests.
- Demonstrated multi-step planning and tool-use by prototyping a multi-agent planner extension that decomposes complex user questions into sub-retrievals and synthesizes real-time results.
- Optimized RAG retrieval relevance by prototyping lightweight intent classification models with TensorFlow and PyTorch to serve

as high-speed pre-retrieval filters.

- Created a fluid, real-time streaming user interface using the Vercel AI SDK coupled with a FastAPI backend, deployed automatically via GitHub Actions to Vercel and Railway.

Vinukonda Kirana: Ecommerce Mobile App

Dec 2025 - Apr 2026

Personal Project

- Shipped a 0-1 React Native + Expo mobile application to production featuring custom client, admin, and delivery roles secured with strict role-based access control (RBAC).
- Automated catalog updates and eliminated manual entry errors by building a Node.js + Firebase Admin SDK data pipeline to ingest, validate, and load massive product catalogs into Firestore.
- Secured critical order-to-payment flows by integrating Razorpay payment gateways with robust webhook confirmation, failure recovery, and real-time state synchronization.
- Expanded target market reach by implementing multilingual localization (English, Telugu, Hindi) via i18next and validated core payment flows using Jest and React Native Testing Library.

EDUCATION

Master of Science, Computer Science

Expected Dec 2026

University of the Cumberlands | Williamsburg, KY

Bachelor of Science, Computer Science

Aug 2022

University of Massachusetts at Lowell | Lowell, MA

CERTIFICATIONS

AWS Certified Cloud Practitioner | Amazon Web Services, Feb 2026

Salesforce Certified Platform Administrator | Sept 2022